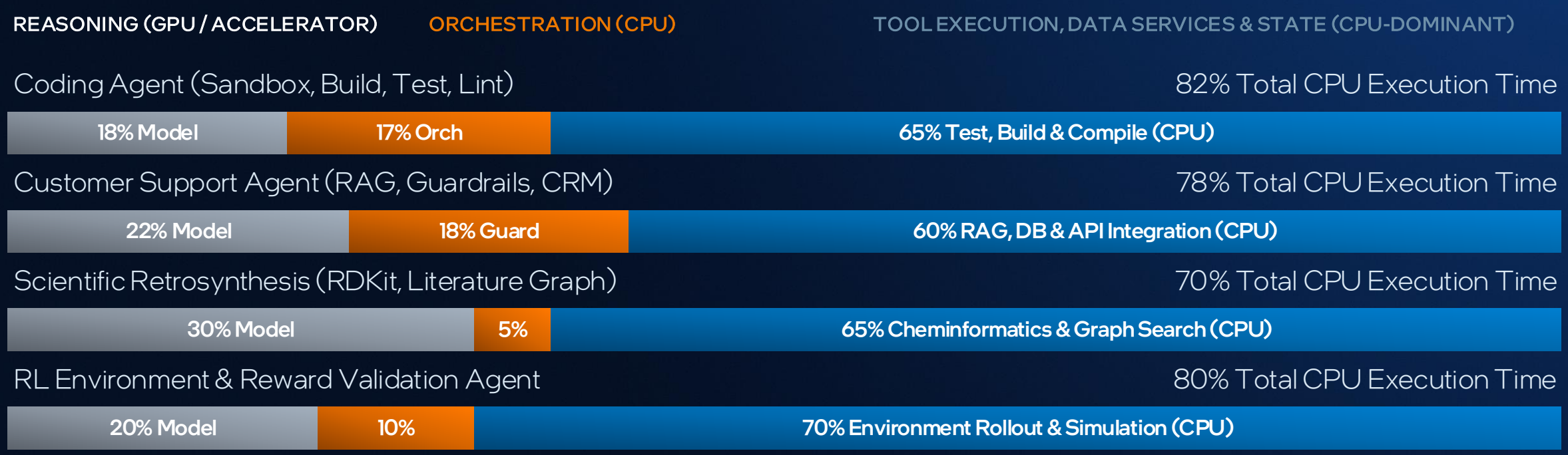
The Agentic Rack, powered by Intel® Xeon® Processors

Jatin Upadhyay
DATA CENTER GROUP, INTEL



Agentic AI Requires a Different Hardware Mix

Where Wall-Clock Latency is Actually Spent*



*illustrative simulation estimate

CPU execution remains critical to end-to-end agent performance

Different Tasks in Agentic Flow Require Different CPU Characteristics

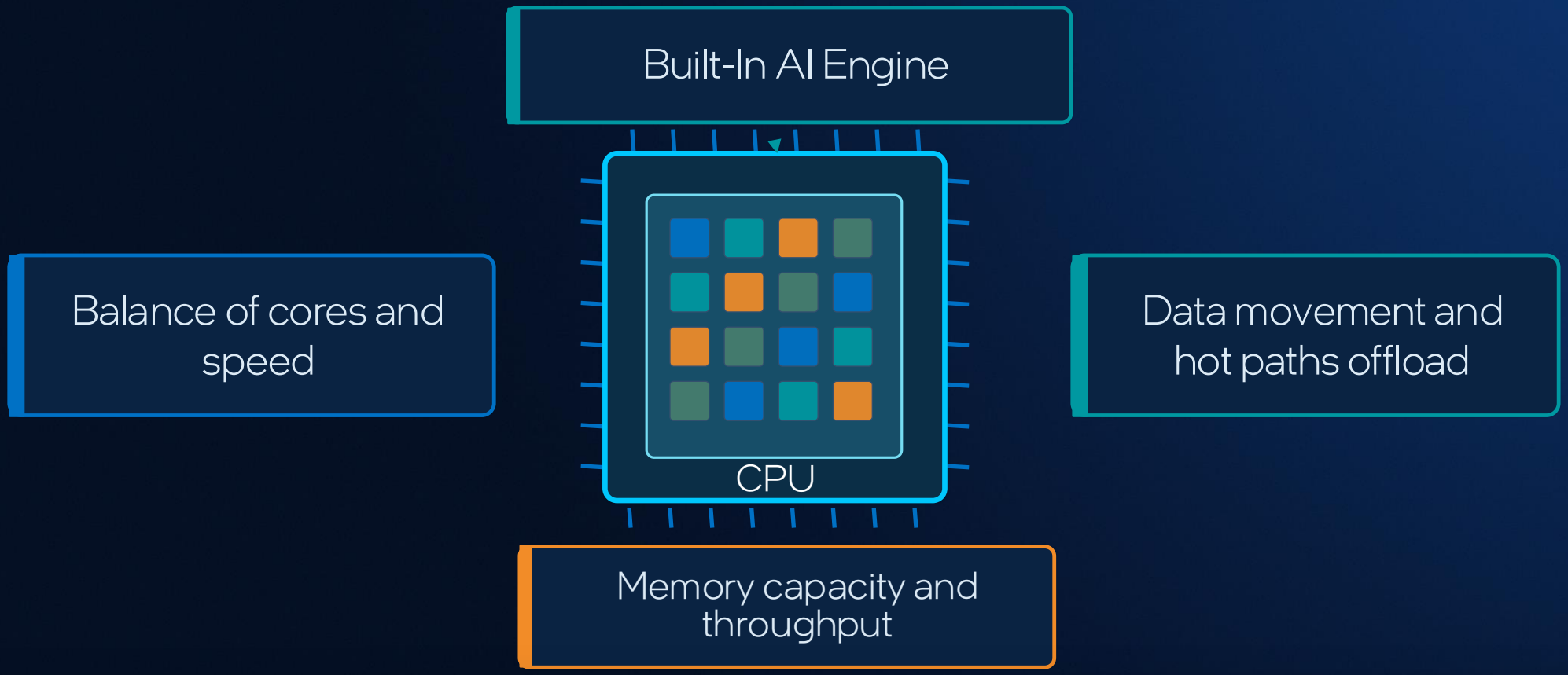
one agent turn, by share of wall-clock



Model call	Orchestration	Tool execution	Data services	State movement
prefill + decode	sandboxes, code, SQL	retrieval, vector, RAG	& scheduling	KV, TLS, tiering
GPU work	*Per-thread speed*	*Core density*	*Memory Performance*	

No single “best” core for every Agentic task;
Flexibility and “balance” of CPU capabilities matter

CPU Capabilities that determine End to End “Agent” Performance



The best Agentic CPU is balanced across compute, data movement, memory and connectivity.

Four Unique Advantages for Agentic AI Powered by Intel® Xeon® Processors

Different “Cores” for different tasks

Performance cores hold the orchestration path; dense cores run the tool swarm.

On-Die AI Engine – Intel ® Advanced Matrix Extensions (Intel® AMX)

Small models in the agent loop, including routers, guardrails and re-rankers, stay on the CPU.

Native CXL Memory Expansion, Memory Compression

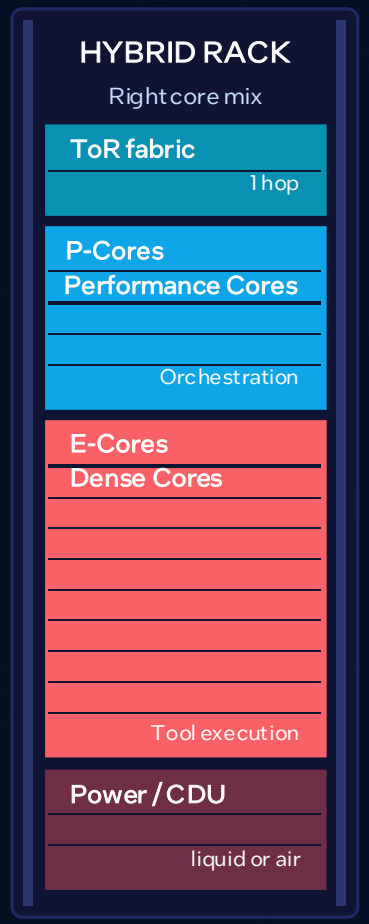
Flat Memory Mode provides hardware-managed memory-side cache over CXL, transparent to the OS. Pooled memory trays expand capacity.

Built-in Data Accelerators

Intel® QuickAssist Technology (Intel® QAT), Intel® Data Streaming Accelerator (Intel® DSA), Intel® In-Memory Analytics Accelerator (Intel® IAA) provide built-in compression, encryption and data movement.

Right core mix. Native AI. Built-in acceleration. Memory capacity.

INTEL ADVANTAGE #1
The right core for every stage of the agent loop



P-cores: accelerate the critical path

Tail-latency sensitive control loop orchestration, branch-heavy policy checks, and coordinator node duties.

E-cores: scale parallel execution

High core density and energy efficiency for concurrent tool sandboxes, code linting, and data parsers.

Hybrid rack | right-sized

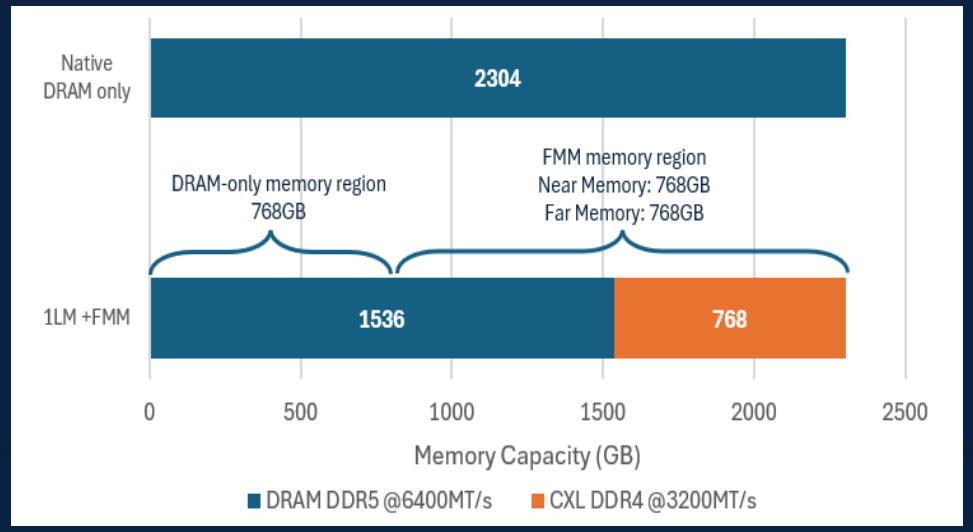
Flexible core mix driven by workload needs

Response time and throughput with same application stack

INTEL ADVANTAGE #2
Hardware-based memory expansion and compression

INTEL® FLAT MEMORY MODE

Hardware-managed CXL memory expansion — one flat address space

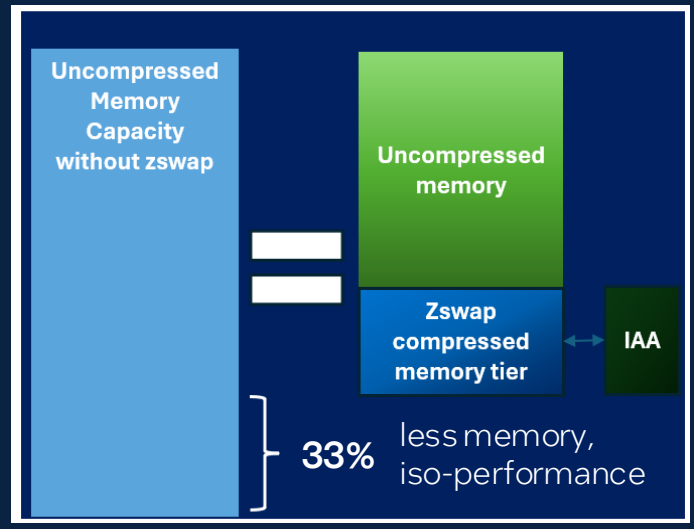


KEY BENEFITS

CXL-attached memory appears as one flat address space. 97.3% of all-DDR5 performance in the cited workload; More resident sessions at lower memory TCO.

IN-MEMORY ANALYTICS ACCELERATOR (Intel® IAA)

On-die compression — on every Intel Xeon CPU



KEY BENEFITS

Compression is handled by the on-die accelerator. No separate BOM line or rack-design change.

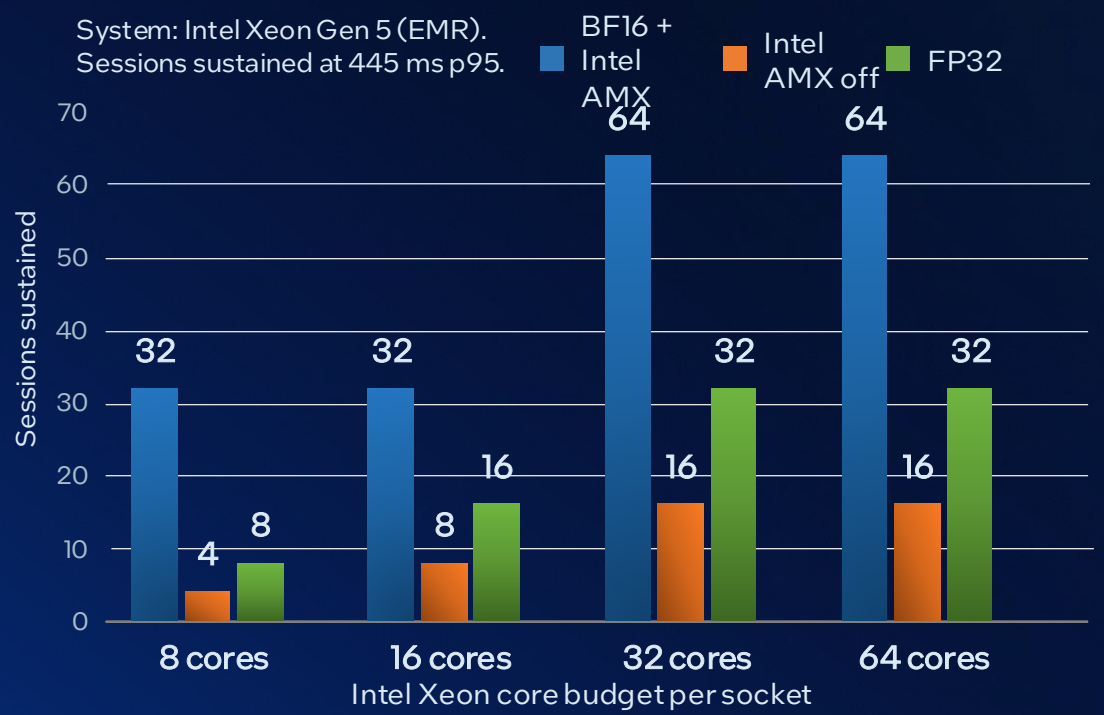
SOURCES AND CONFIGURATION

Flat Memory Mode: 97.3% of all-DDR5 performance on the cited workload. Intel® IAA / zswap: 33% memory reduction at iso-performance. Full configuration disclosure available on request.

Expand capacity with CXL. Reclaim it with a built-in compression. No application change.

INTEL ADVANTAGE #3
Intel® AMX increases agent density and cuts token cost

Semantic Router on Intel Xeon CPUs



Semantic router runs on every agent turn.
BF16 with Intel AMX serves 4x the sessions at the same p95 latency.

With Intel AMX BF16, an Intel Xeon CPU budget sustains up to 4x the sessions than without Intel AMX

SLM inference on Intel Xeon CPUs

Opus 4.8 + Qwen-on-Intel Xeon Vs.
Opus 4.8 + Claude Haiku-in-the-cloud.

Qwen3-Coder-30B-A3B served on an Intel Xeon CPU via matrix acceleration (Intel AMX).

- ✓ 25% lower Token Cost
- ✓ 6.5% lower wall clock time
- ✓ The CPU subagent matched the cloud subagent on every task.

Best Tokenomics: Subagents hosted on Intel Xeon CPUs can cut cost and wall time at equal task outcomes

INTEL ADVANTAGE #4
Convert built-in accelerators into agent density

Intel QAT

Intel QuickAssist Technology

- Hardware cryptographic offload and inline data compression for agent storage, TLS session termination, and secure inter-agent fabrics

Intel DSA

Intel Data Streaming Accelerator

- Dedicated DMA engine for high-throughput memory copies, context marshalling, and zero-copy data transformations across PCIe/CXL.

Intel IAA

Intel In-Memory Analytic Acceleration

- Sandbox hydration for tool execution
- Sabre (OSDI'24): 4.5x snapshot compression, 55% faster memory restoration

Keep general-purpose cores focused on agent outcomes while integrated engines move, compress, encrypt and analyze data.

Intel® Xeon®: Agentic AI Rack Ecosystem

Faster, safer, more predictable, and ready for your workload

ASRock Rack



ORv3 OCP

ASUS



GIGABYTE



MiTAC Computing



ORv3 OCP

msi



SUPERMICRO

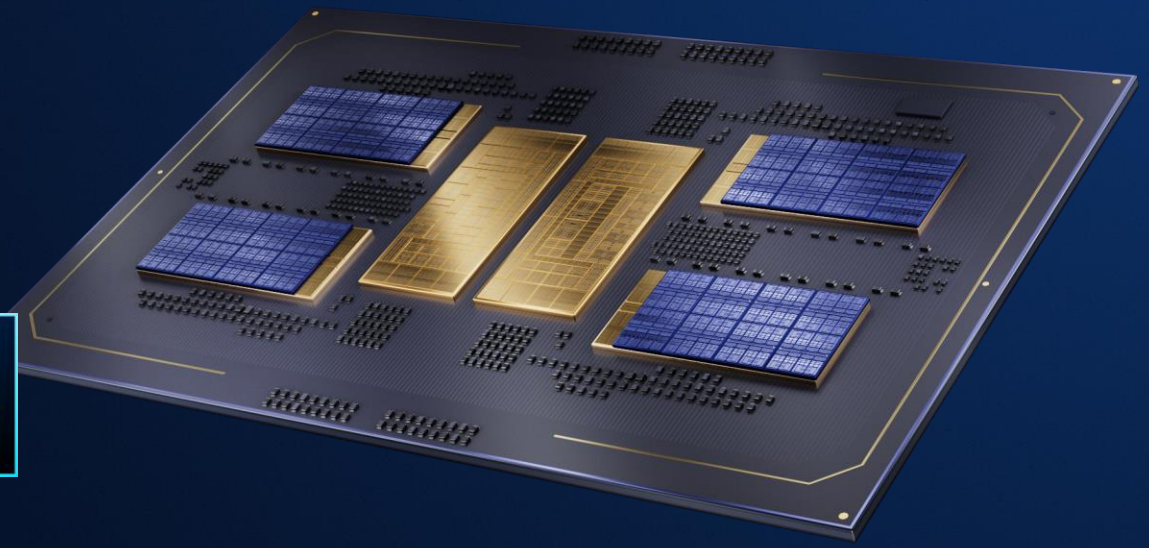


ORv3 OCP

Partners with solutions available today

Performance | Efficiency | Acceleration | Scalability | Security

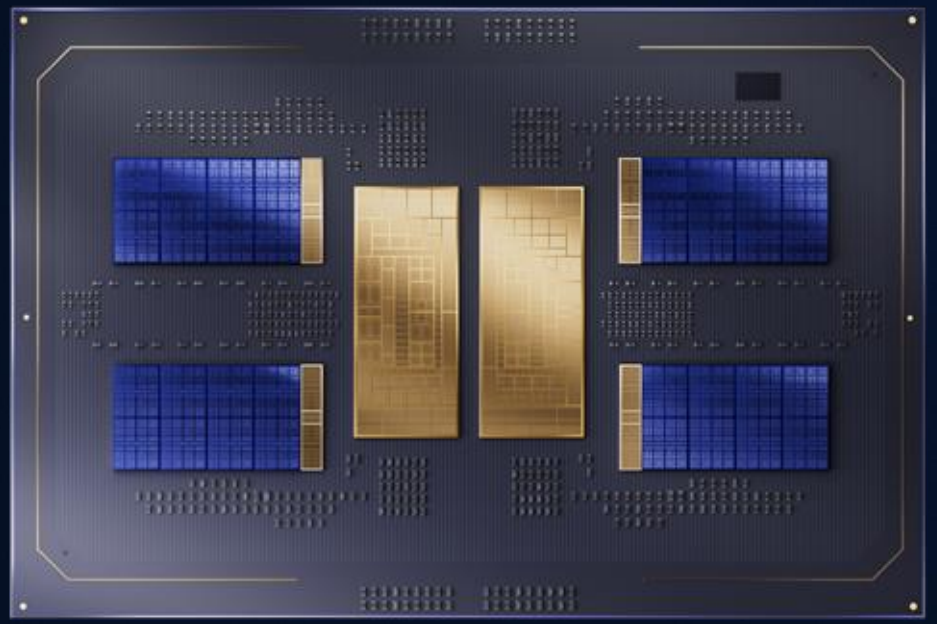
Diamond Rapids: the next-generation Intel Xeon processor



intel 18A-P

High-throughput Compute	Efficient Performance	Infrastructure & AI Acceleration	Modular Architecture	Multi Level Security
Up to 256 cores	Unified Memory Fabric	Intel AMX	Fan-out Fabric Architecture	Intel ® Trust Domain Extensions (Intel ® TDX),
Built on Intel 18A-P	Independent DVFS	Intel QAT, Intel DSA, Intel IAA	Modular CBBs	Improved RAS Enhanced QoS & Telemetry
Massive LLC	Lower Power I/O Fabric		Foveros 3D Direct (HBI)	

Designed for Next Gen Agentic AI Infrastructure



Orchestrate more concurrent work

High compute density for control-plane, tool execution, retries and policy.

Keep data moving

Expanded memory capability and next-generation I/O for context, state and connectivity.

Accelerate and protect

Intel AMX, Intel AVX, Intel QAT, Intel DSA, Intel IAA and platform security as a unified foundation.

256
cores/socket
1.28 GB LLC

16
Memory Channels

32
DIMMs/socket in 2DPC

1.6 TB/s
12800 MT/s

Intel AMX 2.0 & Intel AVX Intel QAT, Intel DSA, Intel IAA

Intel TDX, Post Quantum Cryptography (PQC)

Foundation for orchestration, data movement, acceleration and memory-rich execution

Intel Portfolio of Agentic Solutions

Scale Agentic AI Across the Intelligence and Execution Layers

Edge (CPU-only)

Small models, low concurrency



Simple, compact, cost efficient.

Enterprise Server/Rack (CPU + GPU)

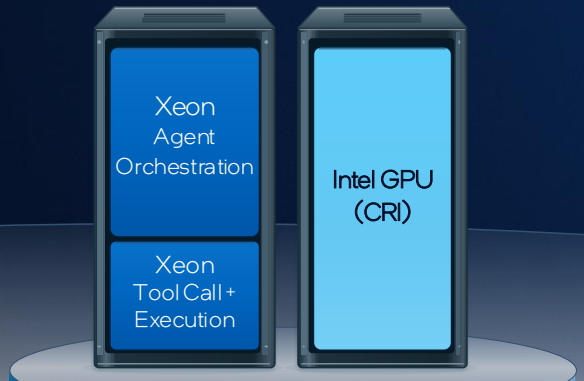
Larger models, higher throughput



Integrated CPU + GPU in one Server and Rack.

Scale-Out Inference and CPU

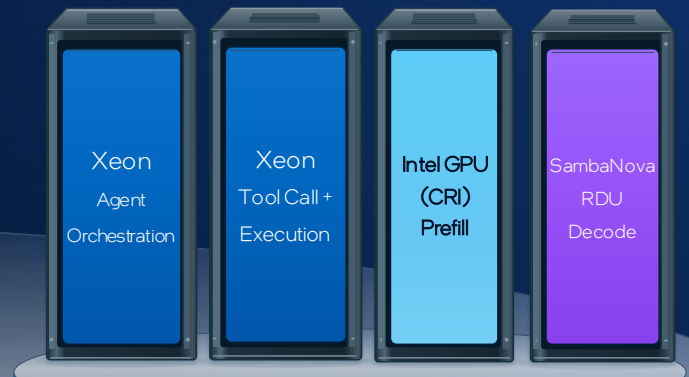
Multiple inference racks + dedicated CPU rack



Scale independently.

Rack-Scale Disaggregation

Disaggregated prefill/decode + dedicated CPU racks



Highest throughput. Fully scalable.

- Open & Ecosystem Ready
- Optimized for Performance and TCO
- Flexible Deployment Options
- Built for What's Next

From Silicon Leadership to Shared Industry Architecture

Intel contributes technology, validation and software; the ecosystem turns it into deployable systems.

INTEL	**Silicon + platform IP**	Intel Xeon / Diamond Rapids(DMR) ▪ Built-in accelerators ▪ Intel RDT ▪ Intel® Application Energy Telemetry (Intel® AET) ▪ Intel TDX ▪ software
ODM / OEM	**Productize + differentiate**	Rack mechanics ▪ cooling ▪ power ▪ serviceability ▪ lifecycle
OCP COMMUNITY	**Standardize + scale**	Open specifications ▪ interfaces ▪ telemetry ▪ manageability
CUSTOMERS	**Co-design around real work**	Workloads ▪ SLOs ▪ TCO ▪ facilities ▪ co-validation

Together we can turn open architecture into right sized, deployable Agentic AI systems

